Standard electrode potential ($E^\ominus$)

Note: The more positive the $E^\ominus$ value, the more favored the forward rxn

$2H^+ + 2e^- \rightleftharpoons H_2$	$E^\ominus = 0.00$ V
$Al^{3+} + 3e^- \rightleftharpoons Al$	$E^\ominus = -1.66$ V

➢ Explain whether H^+ or Al^{3+} has a greater tendency to be reduced.

______ has a greater tendency to undergo the (forward) reduction

because $E^\ominus$(_________) is ________________than $E^\ominus$(_________).

➢ Explain whether is H_2 or Al has a greater tendency to be oxidised.

______ has a greater tendency to undergo the (backwards) oxidation

because $E^\ominus$(_________) is ________________ than $E^\ominus$(_________).

Standard cell potential ($E^\ominus_{cell}$)

An electrochemical cell is setup with two half-cells: H^+ / H_2 *and* Al^{3+} / Al

➢ Write the half equation occurring at the cathode

➢ Write the half equation occurring at the anode

➢ Write the overall cell reaction

➢ Calculate the $E^\ominus_{cell}$ value of the cell setup

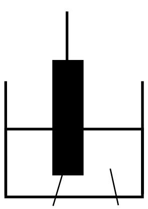
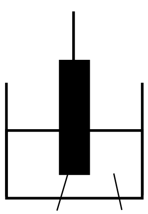
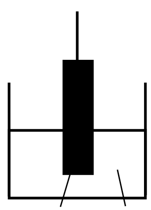
$E^\ominus_{cell} = E^\ominus_{red} - E^\ominus_{ox}$

$=$

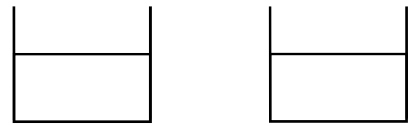
*Note: Half-cell and overall cell equation **must** be written with "$\longrightarrow$"*

Half-cell systems

➢ Draw and label the following sets of half-cells

		
metal/ metal ion (Al^{3+} (aq) / Al (s))	gas/ ion (H^+ (aq) / H_2 (g))	ion/ ion (Fe^{3+} (aq) / Fe^{2+} (aq))

Electrochemical cell setup



half cell 1: H^+ / H_2 half cell 2: Al^{3+} / Al

➢ Draw a labelled electrochemical cell setup to measure the $E^\ominus_{cell}$ consisting of the above 2 half cells.

- electrode
- salt bridge
- voltmeter
- cathode/ anode
- direction of e^- flow
- + / –
- temperature
- pressure
- concentration

<u>Role of the salt bridge</u>

① ensure charge neutrality

② complete the circuit

Effects of POE shift

Note: If forward reaction is favored, *E* value increases. (vice versa)

Data booklet	$2H^+ + 2e^- \rightleftharpoons H_2$ (1)	$E^\ominus = 0.00$ V
	$Al^{3+} + 3e^- \rightleftharpoons Al$ (2)	$E^\ominus = -1.66$ V
Overall eqn	$6H^+ + 2Al \rightarrow 3H_2 + 2Al^{3+}$	$E^\ominus$cell = +1.66 V

➢ Explain how the E_{cell} value be affected when the concentration of Al^{3+}(aq) is increased.

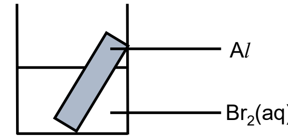
- When [Al^{3+}] increases, position of equilibrium of (2) shifts _______
- The E(Al^{3+}/ Al) becomes __________ _________________
- Hence, E_{cell} becomes __________ _________________

$E^\ominus_{cell}$ and spontaneity

$$\Delta G^\ominus = -nFE^\ominus$$

$E^\ominus_{cell}$	$\Delta G^\ominus$	Spontaneity
– ve		
+ ve		

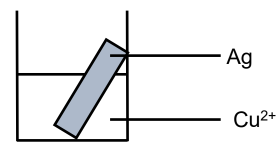
Calculate $\Delta G^\ominus$ and deduce if reaction between Al and Br_2 is spontaneous.



Data booklet	$Br_2 + 2e^- \rightleftharpoons 2Br^-$	$E^\ominus = +1.07$ V
	$Al^{3+} + 3e^- \rightleftharpoons Al$	$E^\ominus = -1.66$ V
[O] half eqn:		$E^\ominus_{cell}$
[R] half eqn:		=
overall eqn:	$2Al + 3Br_2 \rightarrow 2Al^{3+} + 6Br^-$	

−1580 kJ mol⁻¹

Calculate $\Delta G^\ominus$ and deduce if reaction between Ag and Cu^{2+} is spontaneous.



Data booklet	$Ag^+ + e^- \rightleftharpoons Ag$	$E^\ominus = +0.80$ V
	$Cu^{2+} + 2e^- \rightleftharpoons Cu$	$E^\ominus = +0.34$ V
[O] eqn:		$E^\ominus_{cell}$
[R] eqn:		=
Overall eqn:	$2Ag + Cu^{2+} \rightarrow 2Ag^+ + Cu$	

+88.8 kJ mol⁻¹

Standard electrode potential ($E^\ominus$)

Note: The more positive the $E^\ominus$ value, the more favored the forward rxn

$2H^+ + 2e^- \rightleftharpoons H_2$	$E^\ominus$ = 0.00 V
$Al^{3+} + 3e^- \rightleftharpoons Al$	$E^\ominus$ = –1.66 V

➢ Explain whether H^+ or Al^{3+} has a greater tendency to be reduced.

___H^+___ has a greater tendency to undergo the (forward) reduction

because $E^\ominus$(___H^+/H_2___) is ___less negative/ more positive___ than $E^\ominus$(___Al^{3+}/Al___).

➢ Explain whether is H_2 or Al has a greater tendency to be oxidised.

___Al___ has a greater tendency to undergo the (backwards) oxidation

because $E^\ominus$(___Al^{3+}/Al___) is ___more negative___ than $E^\ominus$(___H^+/H_2___).

Standard cell potential ($E^\ominus_{cell}$)

An electrochemical cell is setup with two half-cells: H^+ / H_2 *and* Al^{3+} / Al

➢ Write the half equation occurring at the cathode

(reduction): $2H^+ + 2e^- \longrightarrow H_2$

➢ Write the half equation occurring at the anode

(oxidation): $Al \longrightarrow Al^{3+} + 3e^-$

➢ Write the overall cell reaction

$6H^+ + 2Al \longrightarrow 3H_2 + 2Al^{3+}$

➢ Calculate the $E^\ominus_{cell}$ value of the cell setup

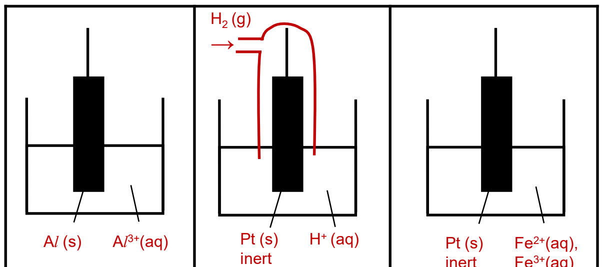
$E^\ominus_{cell} = E^\ominus_{red} - E^\ominus_{ox}$

$= 0.00 - (-1.66) = \underline{+1.66\ V}$

*Note: Half-cell and overall cell equation **must** be written with "⟶"*

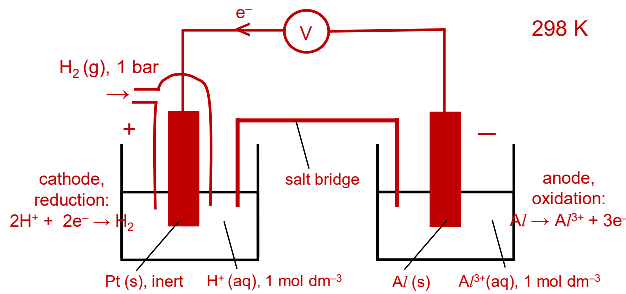
Half-cell systems

➢ Draw and label the following sets of half-cells



metal/ metal ion	gas/ ion	ion/ ion
(Al^{3+} (aq) / Al (s))	(H^+ (aq) / H_2 (g))	(Fe^{3+} (aq) / Fe^{2+} (aq))

Electrochemical cell setup



half cell 1: H^+ / H_2

$2H^+ + 2e^- \longrightarrow H_2$

half cell 2: Al^{3+} / Al

$Al \longrightarrow Al^{3+} + 3e^-$

➢ Draw a labelled electrochemical cell setup to measure the $E^\ominus_{cell}$ consisting of the above 2 half cells.

- electrode
- salt bridge
- voltmeter
- cathode/ anode
- direction of e⁻ flow
- + / –
- temperature
- pressure
- concentration

<u>Role of the salt bridge</u>

1. ensure charge neutrality
2. complete the circuit

Effects of POE shift

Note: If forward reaction is favored, *E* value increases. (vice versa)

Data booklet	$2H^+ + 2e^- \rightleftharpoons H_2$ (1)	$E^\ominus$ = 0.00 V
	$Al^{3+} + 3e^- \rightleftharpoons Al$ (2)	$E^\ominus$ = –1.66 V
Overall eqn	$6H^+ + 2Al \rightarrow 3H_2 + 2Al^{3+}$	$E^\ominus$cell = +1.66 V

➢ Explain how the E_{cell} value be affected when the concentration of Al^{3+}(aq) is increased.

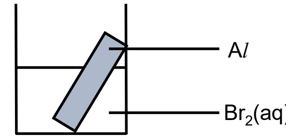
- When [Al^{3+}] increases, position of equilibrium of (2) shifts ___right___
- The $E(Al^{3+}/Al)$ becomes ___less___ ___negative___
- Hence, E_{cell} becomes ___less___ ___positive___

$E^\ominus_{cell}$ and spontaneity

$$\Delta G^\ominus = -nFE^\ominus$$

$E^\ominus_{cell}$	$\Delta G^\ominus$	Spontaneity
– ve	+ ve	not spontaneous
+ ve	– ve	spontaneous

Calculate $\Delta G^\ominus$ and deduce if reaction between Al and Br_2 is spontaneous.



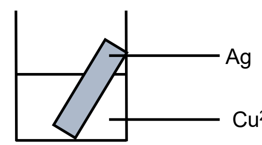
Data booklet	$Br_2 + 2e^- \rightleftharpoons 2Br^-$	$E^\ominus$ = +1.07 V
	$Al^{3+} + 3e^- \rightleftharpoons Al$	$E^\ominus$ = –1.66 V
[O] half eqn:	$Al \rightarrow Al^{3+} + 3e^-$	$E^\ominus_{cell}$
[R] half eqn:	$Br_2 + 2e^- \rightarrow 2Br^-$	= 1.07 – (–1.66)
overall eqn:	$2Al + 3Br_2 \rightarrow 2Al^{3+} + 6Br^-$	= + 2.73 V

$\Delta G^\ominus = -nFE^\ominus$
$= -(6)(96500)(+2.73)$
$= -1580$ kJ mol⁻¹

Since $\Delta G^\ominus$ is negative, reaction is spontaneous

–1580 kJ mol⁻¹

Calculate $\Delta G^\ominus$ and deduce if reaction between Ag and Cu^{2+} is spontaneous.



Data booklet	$Ag^+ + e^- \rightleftharpoons Ag$	$E^\ominus$ = +0.80 V
	$Cu^{2+} + 2e^- \rightleftharpoons Cu$	$E^\ominus$ = +0.34 V
[O] eqn:	$Ag \rightarrow Ag^+ + e^-$	$E^\ominus_{cell}$
[R] eqn:	$Cu^{2+} + 2e^- \rightarrow Cu$	=+0.34 –(+0.80)
Overall eqn:	$2Ag + Cu^{2+} \rightarrow 2Ag^+ + Cu$	= – 0.46 V

$\Delta G^\ominus = -nFE^\ominus$
$= -(2)(96500)(-0.46)$
$= +88.8$ kJ mol⁻¹

Since $\Delta G^\ominus$ is positve, reaction is not spontaneous

+88.8 kJ mol⁻¹